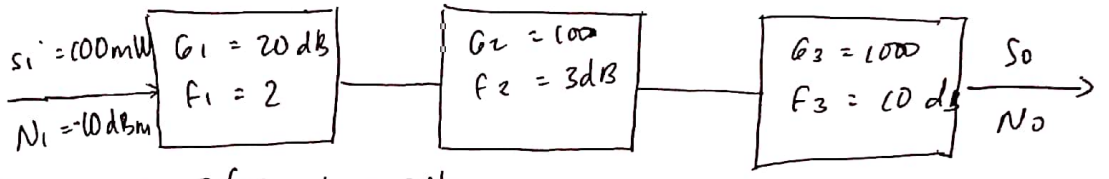
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Kelas : 2NK
UTS Sistem Komunikasi I

بسم الله الرحمن الرحيم

① . Diketahui : penguat bertingkat seperti gambar 1 berikut.



Gambar 1

Hitung: a.) S/N di input
b.) Noise Figure Total
c.) S/N di output.

Jawab.

Diket $S_i = 100\ mW = 100 . 10^{-3}\ W$

$N_i = -10\ dBm = 10^{\frac{-10}{10}} = 10^{-1}\ mW = 0,1 . 10^{-3}$

$G_1 = 20\ dB = 10^2 = 100$

$F_1 = 2$

$G_2 = 100$

$F_2 = 3\ dB = 2$

$G_3 = 1000$

$F_3 = 10\ dB = 10^{10/10} = 10$

Gambar 1

a.) $\frac{S_i}{N_i} = \frac{100\ mW}{0,1\ mW} = 1000$ (ratio)

b.) $F_{t(ratio)} = F_{1\ ratio} + \left(\frac{F_{2\ ratio} - 1}{G_1}\right) + \left(\frac{F_{3\ ratio} - 1}{G_1 . G_2}\right)$

$= 2 + \left(\frac{2-1}{100}\right) + \left(\frac{10-1}{100 . 100}\right)$

$= 2 + 0,01 + 9 . 10^{-4}$

$= 2,0109_{(ratio)} = 10 \log (2,0109) =$

$= 3,03\ dB$

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c.). $G_{Tot} = G_1 \cdot G_2 \cdot G_3$

$= 100 \cdot 100 \cdot 1000$

$= 10^7$ (ratio)

$S_o = G_{tot} \cdot S_i$

$= 10^7 \cdot 100 \cdot 10^{-3}$

$= 10^6$ W

$N_o = G_{tot} \cdot N_i$

$= 10^7 \cdot 0{,}1 \cdot 10^{-3}$

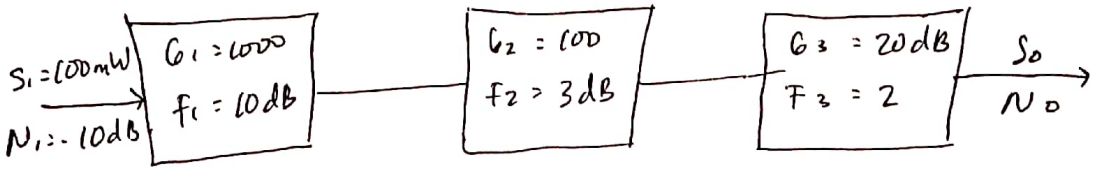
$= 10^3$ W

$\frac{S_o}{N_o} = \frac{10^6}{10^3} = 10^3$ (ratio)

$= 10 \log (10^7)$

$= 30$ dB

Gambar 2



Jawab.

d.). S/N di input.

Diket $S_i = 100$ mW $= 100 \cdot 10^{-3}$ W

$N_i = -10$ dBM $= 0{,}1 \cdot 10^{-3}$ W

$G_1 = 1000$

$F_1 = 10$

$G_2 = 100$

$F_2 = 3$ dB $= 2$

$G_3 = 20$ dB $= 100$

$F_3 = 2$

$\frac{S_i}{N_i} = \frac{100 \text{ mW}}{10^{-1} \text{ mW}} = 1000$ (ratio).

e.). Noise Figure Total.

$F_{t\,(ratio)} = F_{1\,ratio} + \left(\frac{F_{2\,ratio} - 1}{G_1}\right) + \left(\frac{F_{3\,ratio} - 1}{G_1 \cdot G_2}\right)$

$= 10 + \left(\frac{2-1}{1000}\right) + \left(\frac{2-1}{1000 \cdot 100}\right)$

$= 10 + 10^{-3} + 10^{-5}$

$= 10$ (ratio)

$= 10 \log 10$

$= 10$ dB

=.

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f). S/N di output

$G_T = G_1 \cdot G_2 \cdot G_3$
$= 100 \cdot 100 \cdot 100$
$= 10^7$ (ratio)

$S_o = G_T \cdot S_i$
$= 10^7 \cdot 100 \cdot 10^{-3}$
$= 10^6$ W

$N_o = G_T \cdot N_i$
$= 10^7 \cdot 0{,}1 \cdot 10^{-3}$
$= 10^3$ W

$\frac{S_o}{N_o} = \frac{10^6}{10^3} = 10^3$ (ratio)
$= 10 \log 10^3$
$= 30$ dB

h.) Menurut saya pada gambar 1 dan 2 lebih bagus penguat gambar 1 karena memiliki noise figure yang kecil

(2). Diket $P_t = 10$ kW

$V_m(t) = 10 \sin 2\pi\, 5000$ Volt

$F = 100$ KHz

$V_{am}(t) = 20$ V

a). Index modulasi m

$V_{max} = V_c + V_m$
$20 = V_c + 10$
$V_c = 10$
$m = \frac{V_m}{V_c} = \frac{10}{10} = 1$

b). Prosentasi modulasi % m

$\%m = m \cdot 100\%$
$= 1 \cdot 100\%$
$= 100\%$

c) Daya yg digunakan sinyal informasi

$P_t = P_c \left(1 + \frac{m^2}{2}\right)$
$10K = P_c \left(1 + \frac{1^2}{2}\right)$
$P_c = 6666.6$ W

$P_t = P_c + P_m$
$P_m = P_t - P_c$
$= 10k - 6666.6$
$= 3333{,}4$ W

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d.) Daya untuk Carrier

$P_t = P_c \left(1 + \frac{I^2}{2}\right)$

$10k = P_c (1,5)$

$P_c = 6666,6.\ W$

e) $Bw = 105k - 95k$

$= 10\ kHz$

$Bw = 2\ fm = 2.5\ kHz$

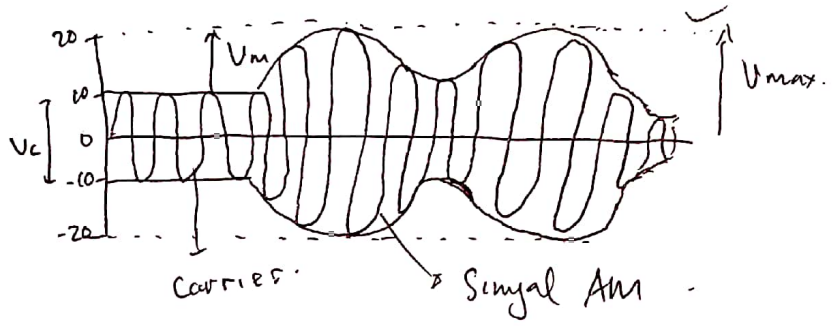
$= 10\ kHz$

f.) $V_c = V_{max} - V_m$

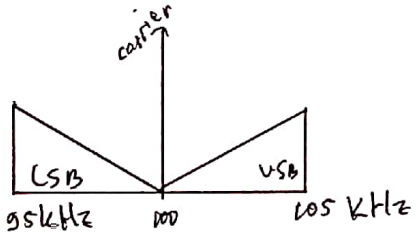
$= 20 - 10$

$= 10\ Volt$

g).



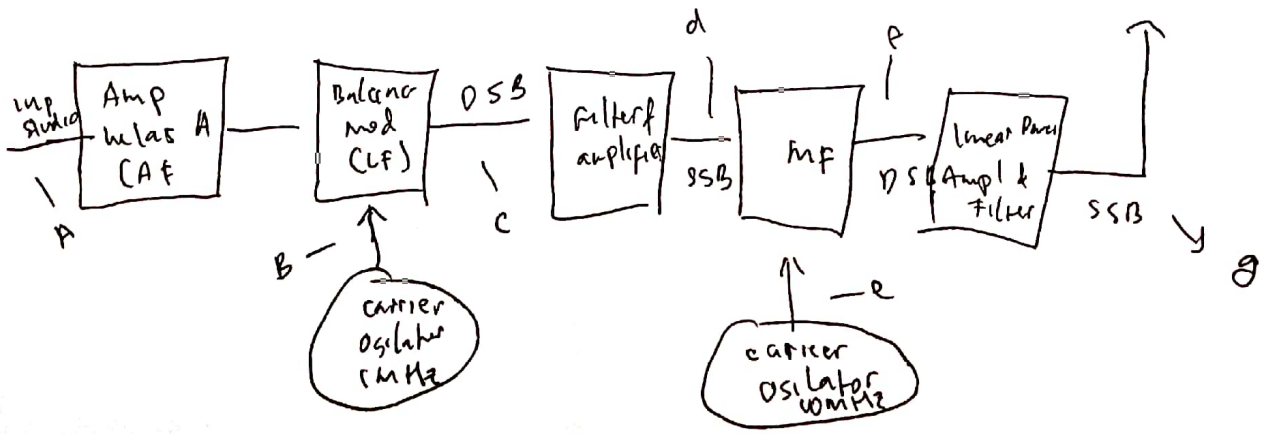
h).



$Bw = 105\ kHz - 95\ kHz$

$= 10\ kHz$

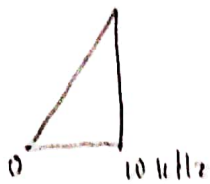
(3) Diketahui diagram blok Pemancar SSB
Gambarkan spektrum sinyal a, b, c, d, e, f, g beri batas frekuensinya



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3)

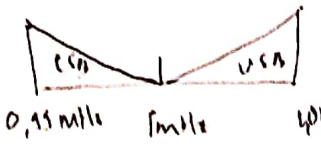
a)



b)



c).



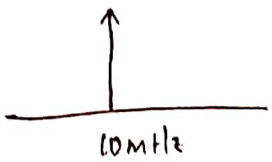
$LSB = 10^6 - 10k$
$= 0,99$

$USB = 10^6 + 10k$
$= 1,01\ mHz$

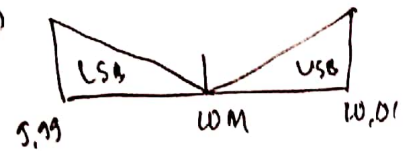
d.)



e).



F.)



$LSB = 10.10^6 - 10.10^3$
$= 9,99\ mHz$

$USB = 10.10^6 + 10.10^3$
$= 10,01\ mHz$

g.)



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